

WASP-2b

R-0145

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_250626 · created 2026-07-16T08:25:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460852.7973 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1396 +/- 0.009
Transit depth (Rp/Rs)^2	1.95 +/- 0.25 [%]
Orbital Inclination (inc)	85.49 +/- 0.61
Airmass coefficient 1 (a1)	1.0072 +/- 0.009
Airmass coefficient 2 (a2)	-0.0045 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.89 %
Best Comparison Star	#3 - [35, 258]
Optimal Aperture	4.04
Optimal Annulus	10.15
Transit Duration (day)	0.0811 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1396 $\pm$ 0.009 vs 0.1326 (deviation 0.66 σ)
Duration fit vs published	0.0811 d vs 0.0737 d (deviation 1.22 σ)
Mid-time O-C	-2.5 min
Depth SNR	13.1
Residual scatter	0.89 %

Light curve

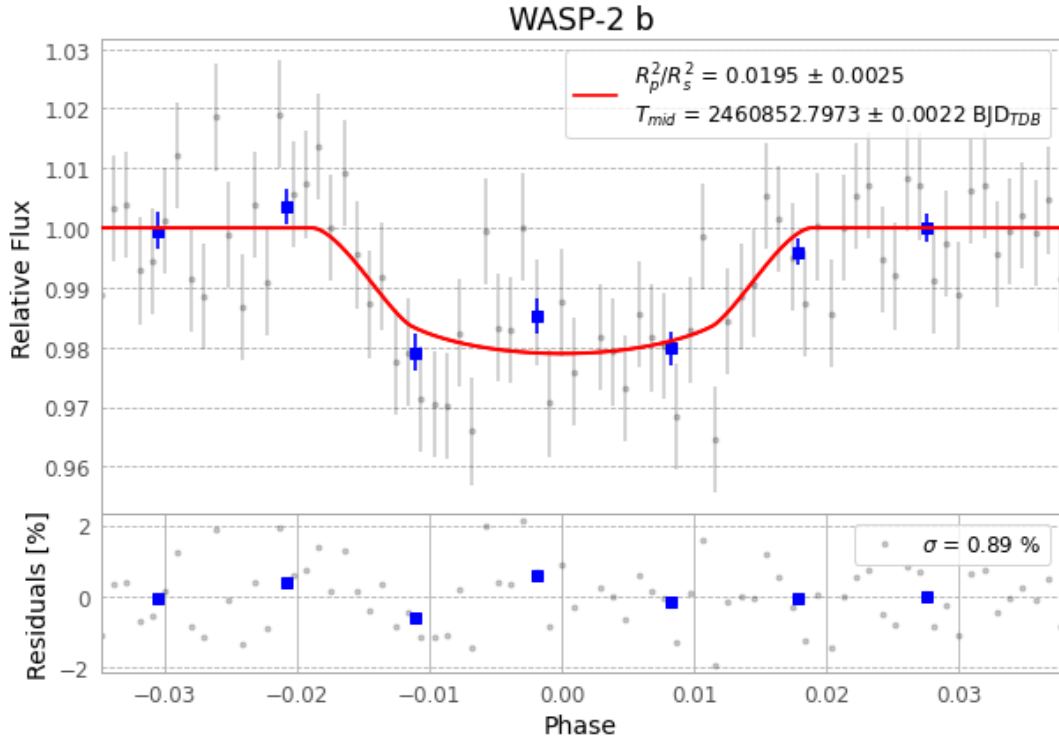


Figure 1 — FinalLightCurve_WASP-2 b_2025-06-25.png (SHA-256 71fbef90aa190c3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc5fe7093d4e2843...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

76 FITS frames (50 MB), WASP-2250626051215.FITS → WASP-2250626085716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250626.log (4964cc717d3e...), FinalLightCurve_WASP-2 b_2025-06-25.png (71fbef90aa19...), FinalLightCurve_WASP-2 b_2025-06-25.csv (53b93495d70c...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 08:25Z.